

习题13.1 p172 1(1,4,6), 2,4,5

习题13.2 p179 1-4中任选3道题, 5-8中任选3道题, 共6道题

习题13.3 p189 1,3, 6-9中任选2道题

习题13.5 p203 奇数中任选3道题, 偶数中任选3道题, 共6道题

习题13.6 p208 所有12道题中中任选6道题

13.1

13.1.1

Determine radius of convergence, and convergence at boundaries.

(1) $\sum \frac{x^n}{n(n+1)}$: $\rho = 1$, convergent at $x = \pm 1$;

(4) $\sum H_n x^n$: $\rho = 1$, diverges at $x = \pm 1$.

(6) $\sum \frac{(\log n)^\alpha}{n^\alpha} x^n$, $\alpha \in \mathbb{R}$: $\rho = 1$, converges at $x = 1$ iff $\alpha > 1$, converges at $x = -1$ iff $\alpha > 0$.

13.1.2

Determine radius of convergence of $\sum \frac{\sin n\alpha}{n} x^n$.

Solution: $\limsup_{n \rightarrow \infty} (\sin n\alpha)^{1/n} = 1$ so $\rho = 1$.

13.1.4

Suppose $l = \liminf_{n \rightarrow \infty} |a_n/a_{n+1}|$, $L = \limsup_{n \rightarrow \infty} |a_n/a_{n+1}|$, and the radius of convergence of $\sum a_n x^n$ is R , prove that $L^{-1} \leq R \leq l^{-1}$.

Proof: Trivial since

$$\liminf_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \leq \liminf_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} |a_n|^{1/n} \leq \limsup_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|.$$

13.1.5

Suppose $\sum a_n x^n$ has radius of convergence 1, and $\lim_{x \rightarrow 1^-} \sum a_n x^n = A$. If $a_n \geq 0$, show that $A = \sum a_n$.

Proof: Clearly $\sum_{n \leq N} a_n = \lim_{x \rightarrow 1^-} \sum_{n \leq N} a_n x^n \leq \lim_{x \rightarrow 1^-} \sum a_n x^n = A$ and $A = \lim_{x \rightarrow 1^-} \sum a_n x^n \leq \sum a_n$. Hence $A = \sum a_n$.

13.2

13.2.3

Suppose $S(x) = \sum a_n x^n$ has radius of converges $R > 0$, prove that

(1) If $\sum \frac{a_n}{n+1} R^{n+1}$ converges then $\int_0^R S(x) dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} R^{n+1}$.

(2) If $\sum n a_n R^{n-1}$ converges, then $S'(R) = \sum n a_n R^{n-1}$.

Proof: (1) By Dominated Convergence Theorem

$$\int_0^R S(x) dx = \lim_{N \rightarrow \infty} \int_0^R \sum_{n=0}^N a_n x^n dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} R^{n+1}.$$

(2) Then $g(x) = \sum n a_n x^{n-1}$ is continuous on $[-R, R]$, so $\frac{S(R)-S(x)}{R-x} = S'(\xi) = g(\xi)$ for some $\xi \in (x, R)$. Let $x \rightarrow R$, then $\lim_{x \rightarrow R} g(x) = S'(R)$ so $S'(R) = \sum n a_n R^{n-1}$.

13.2.4

Suppose $\sum a_n x^n$ converges uniformly on $(-R, R)$. Show that it converges uniformly on $[-R, R]$.

Proof: For any ε there exists N such that $n > m > N$ implies $|\sum_{k=n}^m a_k x^k| < \varepsilon \forall x \in (-R, R)$. Since $\sum_{k=n}^m a_k x^k$ is a polynomial, it is continuous hence $|\sum_{k=n}^m a_k x^k| \leq \varepsilon \forall x \in [-R, R]$, so $\sum a_n x^n$ converges uniformly on $[-R, R]$.

13.2.5

(Du Bois Reymond) Suppose $f \in C([0, 1])$ and $\int_0^1 f(x) x^n dx = 0 \forall n \geq 1$, then $f \equiv 0$.

Proof: For any polynomial $P \in \mathbb{R}[x]$, $\int_0^1 f(x) P(x) dx = 0$. Since $f \in C([0, 1])$ can be approximated by polynomials uniformly, $\int_0^1 f(x)^2 dx = 0$ so $f \equiv 0$.

13.2.6

Suppose a nonconstant f can be expanded as a power series at every $x \in (a, b)$, show that $f^{-1}(0)$ has no limit points in (a, b) .

Proof: Suppose $f(c) = 0$ and $f(x_n) = 0$ where $x_n \rightarrow c$. Then if $f(x) = \sum a_n (x - c)^n$, and a_k is the first non-zero term, for small h we have $|f(c + h)| \geq |a_k| \frac{h^k}{2}$ so f does not vanish in a small neighborhood of c , a contradiction. Hence the power series is identically zero, and $f \equiv 0$ in some neighborhood of c . Therefore the component of $f^{-1}(0)$ containing c is both closed and open, and it is non-empty, so $f^{-1}(0) = (a, b)$ a contradiction.

13.2.7

Suppose f can be expanded as a power series in some neighborhood of $x = 0$, and $f^{(n)}(0)$ is bounded. Show that f is the restriction of some $g \in C^\infty(\mathbb{R})$.

Proof: Let $g(x) = \sum \frac{f^{(n)}(0)}{n!} x^n$ then $g \in C^\infty(\mathbb{R})$. Since $f^{(n)}(0)$ is bounded, $R_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k$ converges to 0 uniformly in any compact set, so $f \equiv g$ locally. By Ex13.2.6 f is the restriction of g to its domain.

13.2.8

Prove that the Bernstein polynomial converges uniformly to f for $f \in C([0, 1])$.

Proof: Note that $\sum_{k=0}^n f(x) \binom{n}{k} x^k (1-x)^{n-k} = f(x)$, and $B_{f,n} = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$. Let $p_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}$, then $\sum_{0 \leq k \leq n} p_{n,k}(x) \left(x - \frac{k}{n}\right)^2 = \frac{x(1-x)}{n}$.

Hence

$$|f(x) - B_{f,n}(x)| \leq \sum_{k=0}^n \left| f(x) - f\left(\frac{k}{n}\right) \right| p_{n,k}(x).$$

Let

$$\begin{aligned} \text{I} &= \sum_{|m-nx| < n^{3/4}} \left(f(x) - f\left(\frac{m}{n}\right) \right) p_{n,m}(x), \\ \text{II} &= \sum_{|m-nx| > n^{3/4}} \left(f(x) - f\left(\frac{m}{n}\right) \right) p_{n,m}(x). \end{aligned}$$

Then $|f - B_{f,n}| \leq |\text{I}| + |\text{II}|$.

For any $\varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall x \in [0, 1], n \geq N \implies |\text{I}| < \varepsilon$, since

$$|\text{I}| \leq \sup_{|x-m/n| < n^{-1/4}} |f(x) - f(m/n)| \rightarrow 0.$$

Suppose $M = \sup_{x \in [0,1]} |f(x)|$, then

$$|\text{II}| \leq 2M \sum_{|m-nx| > n^{3/4}} p_{n,m}(x) \leq 2M\sqrt{n} \sum_{m=0}^n (x - m/n)^2 p_{n,m}(x) = \frac{2Mx(1-x)}{\sqrt{n}}.$$

Hence $\|f - B_{f,n}\|_{\infty} \rightarrow 0$.

13.3

13.3.1

Calculate $\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+1}}{4n+1}$.

Solution:

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+1}}{4n+1} = \frac{\sqrt{2}}{4} \arctan\left(\frac{x\sqrt{2}}{1-x^2}\right) + \frac{\sqrt{2}}{8} \ln\left(\frac{x^2 + x\sqrt{2} + 1}{x^2 - x\sqrt{2} + 1}\right)$$

13.3.6

Calculate

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{3k+2}.$$

Solution: Note that

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{3k+2} = \sum_{k=0}^{\infty} \frac{1}{3k+1} - \frac{1}{3k+2} = L(1, \chi_3).$$

where $\chi_3 : (\mathbb{Z}/3\mathbb{Z})^\times \hookrightarrow \mathbb{C}^*$. Recall for odd χ ,

$$L(1, \chi) = \frac{\pi}{2q} \sum_{k=0}^{q-1} \chi(k) \cot \frac{k\pi}{q} = \frac{\pi}{6} \left(\cot \frac{\pi}{3} - \cot \frac{2\pi}{3} \right) = \frac{2\sqrt{3}}{9} \pi.$$

13.3.7

Calculate

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{4k+1}.$$

Solution: $(\mathbb{Z}/8)^\times = \{1, 3, 5, 7\}$. Consider $\chi : (1, 3, 5, 7) \mapsto (1, -1, -1, 1)$ and $\chi' : (1, 3, 5, 7) \mapsto (1, 1, -1, -1)$ then

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{4k+1} = \frac{1}{2} (L(1, \chi) + L(1, \chi')).$$

and likewise calculate $L(1, \chi') = \frac{\sqrt{2}\pi}{4}$, $L(1, \chi) = -\frac{\tau(\chi)}{q} \sum_{k=1}^{q-1} \log \sin \frac{k\pi}{q} = \frac{\sqrt{2}}{2} \log(1 + \sqrt{2})$. Therefore

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{4k+1} = \frac{\sqrt{2}\pi}{8} + \frac{\sqrt{2}}{4} \log(1 + \sqrt{2}).$$

13.5

13.5.4

Is there an elementary function with convergence radius 0?

Solution: No. Any elementary function is analytic whenever well-defined, so the finite operation of these functions are still analytic.

13.5.6

If $\sum a_n$ is Cesaro summable, then $a_n/(n+1)$ is bounded.

Proof: $\sum a_n$ is Cesaro summable implies $\sigma_n = \frac{1}{n} \sum (n-k)a_k$ converges. Note that $a_n = (n+1)\sigma_n - 2n\sigma_{n-1} + (n-1)\sigma_{n-2}$, so $\left| \frac{a_n}{n+1} \right| \leq |\sigma_n - \sigma_{n-1}| + |\sigma_{n-1} - \sigma_{n-2}| \rightarrow 0$. Hence it is bounded.

13.5.7

Calculate $\sum_{n=1}^{\infty} \frac{2}{4n-1} x^{2n}$.

Solution: Let $|x| = y^2$ then

$$\sum_{n=1}^{\infty} \frac{2}{4n-1} y^{4n} = y \sum_{n=1}^{\infty} 2 \int_0^y t^{4n-2} dt = 2y \int_0^y \frac{t^2}{1-t^4} dt = \frac{y}{2} \ln \left(\frac{1+y}{1-y} \right) - y \arctan(y).$$

13.5.8

Suppose $\sum a_n$ and $\sum b_n$ are both Cesaro summable, is their product $\sum c_n$ Cesaro summable? ($c_n = \sum a_k b_{n-k}$).

Solution: No. For example, consider $a_n = b_n = (-1)^n$, then clearly both are Cesàro summable, but $c_n = (-1)^n(n+1)$ so it is not Cesàro summable.

13.5.9

Suppose $\sum a_n$ and $\sum b_n$ are both Abel summable, prove that $\sum c_n$ is Abel summable, where $c_n = \sum a_k b_{n-k}$.

Proof: Recall we have the product formula $(\sum a_n x^n)(\sum b_n x^n) = \sum c_n x^n$. Hence $\lim_{x \rightarrow 1^-} \sum c_n x^n = \lim_{x \rightarrow 1^-} \sum a_n x^n \sum b_n x^n$ exists, so $\sum c_n$ is Abel summable.

13.5.10

Suppose $\sum a_n$ are Abel summable, show that $\sum a_n$ converges iff

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n k a_k = 0.$$

Proof: If $\sum a_n$ converges then it is Cesàro summable, so

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n k a_k = \sum_{k=1}^n a_k - \frac{1}{n} \sum_{k=1}^n (n-k) a_k = S - S = 0.$$

Conversely, if $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n k a_k = 0$, then let $b_n = \frac{1}{n} \sum_{k=1}^n k a_k$, we have $n b_n - (n-1) b_{n-1} = n a_n$, so $a_n = b_n - (1 - \frac{1}{n}) b_{n-1}$, and the partial sum $S_n = \sum_{k=1}^n a_k = \sum_{k=1}^n \frac{1}{k} b_{k-1}$. Then $\frac{b_{k-1}}{k} = o(k)$ and it is Abel summable, so by Tauber's theorem $\lim_{x \rightarrow 1^-} \sum \frac{1}{k} b_{k-1} x^k = \sum \frac{1}{k} b_{k-1}$. Therefore S_n converges.

13.6

13.6.2

Suppose $f \in C([0, 1])$, prove that

$$\lim_{n \rightarrow \infty} \frac{1}{2^n} \sum_{k=0}^n (-1)^k \binom{n}{k} f\left(\frac{k}{n}\right) = 0.$$

Proof: Take polynomials P_n such that $\|P_n - f\|_\infty < 2^{-n}$, then for $N > \deg P_n$,

$$\frac{1}{2^N} \sum_{k=0}^N (-1)^k \binom{N}{k} P_n\left(\frac{k}{N}\right) = 0$$

so $\lim_{n \rightarrow \infty} \frac{1}{2^n} \sum_{k=0}^n (-1)^k \binom{n}{k} f\left(\frac{k}{n}\right) = 0$.

13.6.3

Suppose $f \in C([a, b])$, prove that if $\int_a^b f(x) x^n dx = 0$ then $f \equiv 0$.

Proof: Take polynomials P_n such that $\|P_n - f\|_\infty \rightarrow 0$, then $\int_a^b f P_n dx = 0$ implies $\int_a^b f^2 dx = 0$, so $f \equiv 0$.

13.6.4

Suppose $f \in C(\mathbb{R})$ can be uniformly approximated by polynomials, show that f is a polynomial.

Proof: Note that if P, Q are polynomials and $P - Q$ is bounded, then they differ only by a constant. Hence $P + c_n \rightarrow f$ for some polynomial P and constants c_n , so $f = P + c$ is a polynomial where $c = \lim_{n \rightarrow \infty} c_n$.

13.6.9

Suppose f is uniformly approximated by polynomials on (a, b) , show that f is uniformly continuous.

Proof: Suppose $\|P_n - f\|_\infty < 2^{-n}$ for polynomials P_n , then

$$|f(x) - f(y)| \leq |f(x) - P_n(x)| + |P_n(x) - P_n(y)| + |P_n(y) - f(y)| \leq 2^{-n-1} + |P_n(x) - P_n(y)|.$$

For any $\varepsilon > 0$, take n such that $2^{-n-1} < \varepsilon/2$, and since P_n is uniformly continuous on $[a, b]$, take $\delta > 0$ such that $|x - y| < \delta \implies |P_n(x) - P_n(y)| < \varepsilon/2$. Therefore $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$, so f is uniformly continuous.

13.6.10

Suppose $f \in C([1, \infty))$ and $\lim_{x \rightarrow \infty} f(x) = c$. Show that for any $\varepsilon > 0$ there is a polynomial such that $\|f(x) - P(1/x)\|_\infty < \varepsilon$.

Proof: Apply Stone-Weierstrass to the function $f\left(\frac{1}{x}\right)$ on $[0, 1]$.

13.6.10

Suppose $f \in C([0, \infty))$ and $\lim_{x \rightarrow \infty} f(x) = c$. Show that for any $\varepsilon > 0$ there is a polynomial such that $\|f(x) - P(e^{-x})\|_\infty < \varepsilon$.

Proof: Apply Stone-Weierstrass to the function $f(-\log x)$ on $[0, 1]$.